# 9.2.2

EE25BTECH11042 - Nipun Dasari

**Question**:
Find the points of intersection of the following parabola and circle and plot the region

$$y^2 \leq 6ax \tag{0.1}$$
$$x^2 + y^2 \leq 16a^2 \tag{0.2}$$

**Solution:**
We first express the equations for the parabola (0.1) and the circle (0.2) in the standard quadratic form $g(x, y) = 0$.

$$y^2 - 6ax = 0 \tag{0.3}$$
$$x^2 + y^2 - 16a^2 = 0 \tag{0.4}$$

In homogeneous coordinates, using the form $\mathbf{x}^\top C\mathbf{x} = 0$, where $\mathbf{x} = \begin{pmatrix} x & y & 1 \end{pmatrix}^\top$, the matrices for the conics are:
where,

$$C = \begin{pmatrix} \mathbf{V} & \mathbf{u} \\ \mathbf{u}^\top & f \end{pmatrix} \tag{0.5}$$

$$C_1 = \begin{pmatrix} 0 & 0 & -3a \\ 0 & 1 & 0 \\ -3a & 0 & 0 \end{pmatrix} \tag{0.6}$$

$$C_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -16a^2 \end{pmatrix} \tag{0.7}$$

The intersection points lie on a degenerate conic formed from a linear combination $P_1 + \mu C_2 = 0$. We must find the value of $\mu$ that makes the determinant of the combined conic's matrix, $C(\mu) = C_1 + \mu C_2$, equal to zero. The general matrix for this combination is:

$$C(\mu) = \begin{pmatrix} \mu & 0 & -3a \\ 0 & 1+\mu & 0 \\ -3a & 0 & -16a^2\mu \end{pmatrix} \tag{0.8}$$

We set the determinant to zero:

$$\begin{aligned} \det(C(\mu)) &= \mu(1+\mu)(-16a^2\mu) - (-3a)(1+\mu)(-3a) \\ &= -16a^2\mu^2(1+\mu) - 9a^2(1+\mu) \\ &= -a^2(1+\mu)(16\mu^2 + 9) = 0 \end{aligned} \tag{0.9}$$

The only real root is obtained from the first factor, which gives $\mu = -1$. The equation for this degenerate conic is found by subtracting (0.4) from (0.3):

$$
\begin{aligned}
(y^2 - 6ax) - (x^2 + y^2 - 16a^2) &= 0 \\
-x^2 - 6ax + 16a^2 &= 0 \\
x^2 + 6ax - 16a^2 &= 0 \\
\implies (x + 8a)(x - 2a) &= 0
\end{aligned} \tag{0.10}
$$

This degenerate conic consists of two vertical lines, $x = 2a$ and $x = -8a$. The line $x = -8a$ does not intersect the parabola $y^2 = 6ax$ in real points (assuming $a > 0$). We thus consider the line $x = 2a$.

We find the intersection of the line $x - 2a = 0$ with the parabola $P_1$ using the formula $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$. The line and conic parameters from $P_1$ are: $\mathbf{h} = \begin{pmatrix} 2a \\ 0 \end{pmatrix}$, $\mathbf{m} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $\mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, $\mathbf{u} = \begin{pmatrix} -3a \\ 0 \end{pmatrix}$, and $f = 0$.

Next, we compute the components for the parameter $\kappa_i$:

$$\mathbf{m}^\top \mathbf{V} \mathbf{m} = 1 \tag{0.11}$$

$$g(\mathbf{h}) = -12a^2 \tag{0.12}$$

$$\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) = 0 \tag{0.13}$$

We substitute these into the formula for $\kappa_i$:

$$
\begin{aligned}
\kappa_i &= \frac{1}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \left( -\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h})(\mathbf{m}^\top \mathbf{V} \mathbf{m})} \right) \\
&= \frac{1}{1} \left( -(0) \pm \sqrt{(0)^2 - (-12a^2)(1)} \right) = \pm \sqrt{12a^2} = \pm 2\sqrt{3}a
\end{aligned} \tag{0.14}
$$

This gives two values: $\kappa_1 = 2\sqrt{3}a$ and $\kappa_2 = -2\sqrt{3}a$.

Finally, we find the intersection points using $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$:

$$\mathbf{x}_1 = \begin{pmatrix} 2a \\ 0 \end{pmatrix} + 2\sqrt{3}a \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2a \\ 2\sqrt{3}a \end{pmatrix} \tag{0.15}$$

$$\mathbf{x}_2 = \begin{pmatrix} 2a \\ 0 \end{pmatrix} - 2\sqrt{3}a \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2a \\ -2\sqrt{3}a \end{pmatrix} \tag{0.16}$$

Thus, the intersection points are $(2a, 2\sqrt{3}a)$ from (0.15) and $(2a, -2\sqrt{3}a)$ from (0.16).

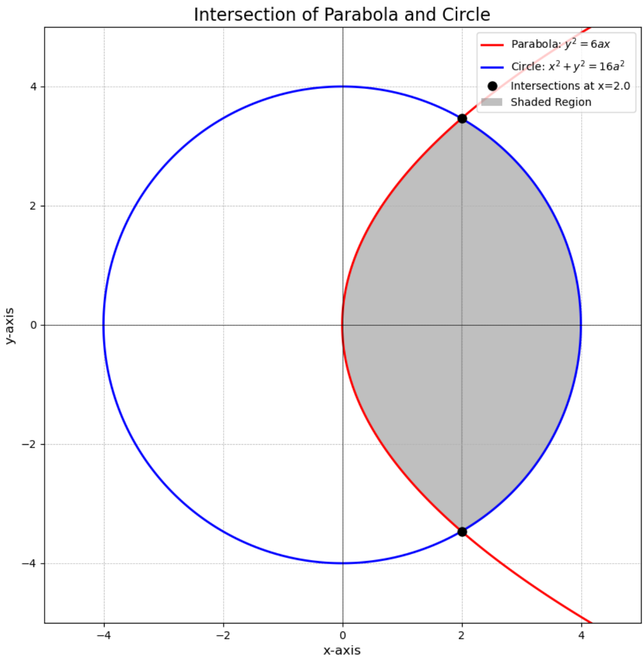


Fig : Parabola and Circle

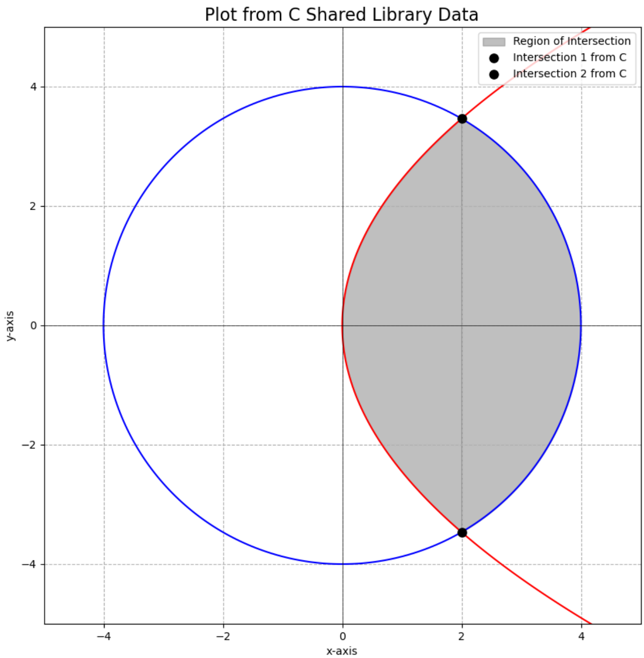


Fig : Intersection Points